EBU5376 Digital Signal Processing

LAB 2

Before you start, please read the following information carefully:

General Instructions:

- The .zip file you downloaded from QMPlus should contain the following documents:
  - This lab sheet (needs to be completed)
  - A MATLAB data file that you need to complete the tasks
- Type your answers in the textboxes provided. Figures need to be copied into the textboxes. If you need more space, enlarge the textbox or add a new box.
- Answers outside the textboxes will NOT be marked.
- If you require any technical assistance, please attend the lab session and ask the on-site TA for help.

How to submit:

- Submit one .zip file to QMPlus. The .zip file should contain:
  1. This LAB SHEET converted to a PDF document
  2. The MATLAB script containing all the work you did

Additional grading criteria:

- Your answers in this sheet must include sufficient explanations/derivations. Marks will be deducted if no explanation is given to support the answer.
- Snapshots of AI-generated results will be ignored.
- Your MATLAB script will be checked from the following three aspects:
  - Whether it can **REPRODUCE ALL THE FIGURES** in this sheet
  - Whether you have added appropriate **COMMENTS** to explain the codes
  - Plagiarism
- If the MATLAB script is not submitted, your grade will be capped to 60%.
- Email submission NOT accepted.
- Do NOT plagiarize. Both the MATLAB script and this sheet will be checked for plagiarism.

Question 1 – Filter Response Analysis **[50 marks]**

In this question, you will practice how to understand filter coefficients by visualising and comparing filter responses.

In Lab 1 materials, you are given a MATLAB data file that consists of the filter coefficients of FOUR different filters. For example, the coefficient arrays of **Filter 1** are named $\boldsymbol{a1}$ (output coefficients) and $\boldsymbol{b1}$ (input coefficients), which corresponds to the transfer function coefficients in

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{\sum_{k=0}^{N-1} a_k z^{-k}}$$

Similarly, the coefficients of **Filter 2,3,4** are named $\boldsymbol{a2}, \boldsymbol{a3}, \boldsymbol{a4}$ (output coefficients) and $\boldsymbol{b2}, \boldsymbol{b3}, \boldsymbol{b4}$ (input coefficients).

Task 1: Among the four filters, identify the Finite Impulse Response (FIR) filters and the Infinite Impulse Response (IIR) filters. Explain your answers.

[5 marks]

To identify FIR (Finite Impulse Response) and IIR (Infinite Impulse Response) filters, we examine the denominator coefficients 'a' for each filter. The transfer function of a digital filter is given by H(z) = B(z)/A(z), where A(z) = ∑ a_k z^{-k} (from k=0 to N-1) and B(z) = ∑ b_k z^{-k} (from k=0 to M-1).

FIR filters have an impulse response of finite duration. Their transfer function has poles only at the origin (z=0), meaning the denominator polynomial A(z) is a constant, typically normalized to 1. Thus, for an FIR filter, the coefficient vector 'a' will be a = [1] or a = [a0] (a single non-zero element, which is a_0). If other elements exist, they are zero (e.g. a = [a0 0 0]). More generally, for an FIR filter, a_k = 0 for k > 0, meaning that if length(a) > 1, all elements a(2:end) must be zero.

IIR filters have an impulse response that is theoretically infinite in duration (due to feedback). Their transfer function has poles at locations other than the origin, meaning the denominator polynomial A(z) is of an order greater than zero with respect to z^{-1}. Thus, for an IIR filter, the coefficient vector 'a' will have at least one non-zero coefficient a_k for k > 0.

After loading the filter_coefficients.mat file and inspecting the 'a' coefficients:

Filter 1 (a1): The coefficient vector a1 is [1.0000 -2.9754 3.8060 -2.5453 0.8811 -0.1254].

Based on this, Filter 1 is an IIR filter because a1 has multiple non-zero coefficients beyond the first element (a1(1)), indicating a non-trivial denominator polynomial A(z) and thus feedback.

Filter 2 (a2): The coefficient vector a2 is [1.0000 -2.9775 4.2932 -3.5124 1.6145 -0.3334].

Based on this, Filter 2 is an IIR filter because a2 has multiple non-zero coefficients beyond the first element, indicating a non-trivial denominator polynomial A(z).

Filter 3 (a3): The coefficient vector a3 is [1.0000 -2.9507 3.7630 -2.5056 0.8649 -0.1225].

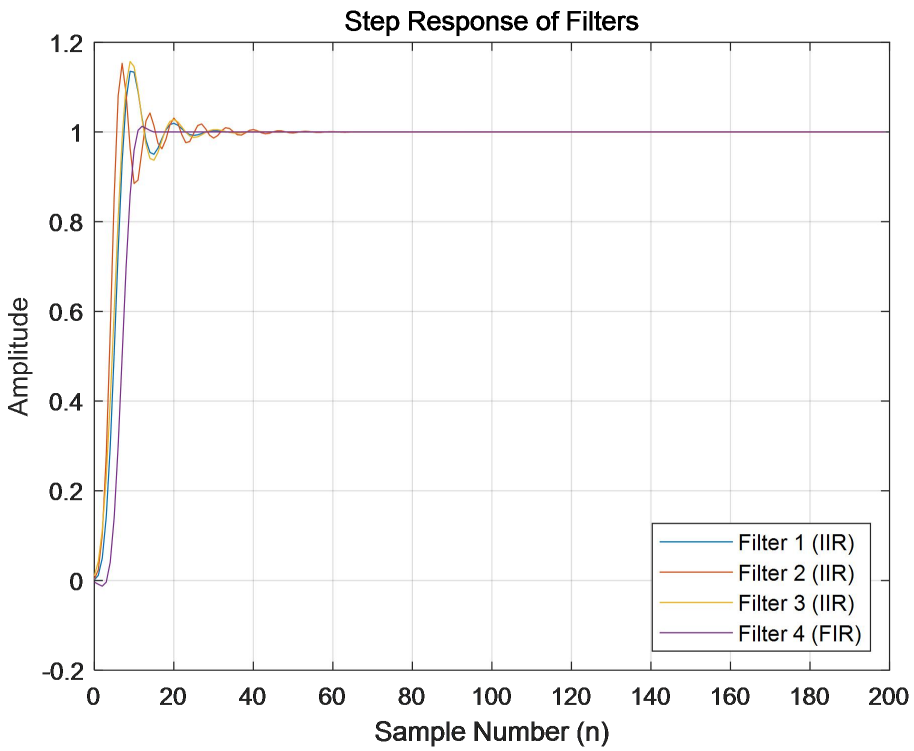
Based on this, Filter 3 is an IIR filter because a3 has multiple non-zero coefficients beyond the first element, indicating a non-trivial denominator polynomial A(z).

Filter 4 (a4): The coefficient vector a4 is [1].

Based on this, Filter 4 is an FIR filter because a4 consists of a single element which is 1. This means the denominator polynomial A(z) = 1, which is characteristic of an FIR filter (all poles are at the origin).

Task 2: Plot the step response of all filters on the same figure. Briefly compare the filters based on the plot.

[10 marks]



The figure above shows the step responses of the four filters plotted over 200 samples. This allows for an observation of both the transient behavior and the settling towards the steady-state amplitude, which is 1.0000 for all filters.

Brief comparison based on the plot and observed characteristics:

Filter 1 (IIR): This filter exhibits a rapid rise, with a 10-90% rise time of 4 samples. It has a significant overshoot, reaching a peak value of 1.1353 (13.53% overshoot). Following the peak, it shows oscillations before settling within $\pm 2\%$ of the steady-state value from sample n=17 onwards.

Filter 2 (IIR): Filter 2 demonstrates the fastest 10-90% rise time of 4 samples. It presents a notable overshoot, peaking at 1.1530 (15.30% overshoot). This is followed by pronounced oscillations. The filter settles within $\pm 2\%$ of the steady-state value from sample n=25.

Filter 3 (IIR): This filter also has a fast rise, with a 10-90% rise time of 5 samples. It shows the largest peak overshoot among the four filters, reaching 1.1570 (15.70% overshoot). Oscillations are apparent before the filter settles within $\pm 2\%$ of its steady-state value from sample n=22.

Filter 4 (FIR): The FIR filter has a 10-90% rise time of 5 samples. Visually, its response is much smoother with significantly less overshoot compared to the IIR filters; the calculated peak value is 1.0126 (1.26% overshoot). It achieves the fastest settling time, remaining within $\pm 2\%$ of the steady-state value from sample n=11 onwards without prolonged ringing.

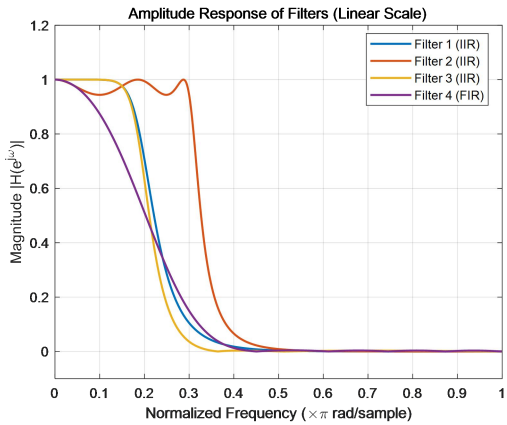
Overall Observations:

All three IIR filters (Filters 1, 2, and 3) display fast rise times but are characterized by considerable overshoot and subsequent ringing (oscillations) as seen in their transient response. Filter 3 has the highest percentage overshoot (15.70%), closely followed by Filter 2 (15.30%), while Filter 1 has a slightly lower overshoot (13.53%).

In contrast, Filter 4 (FIR) provides a much more damped and smoother response with minimal overshoot (1.26%). It is also the quickest to settle within a tight tolerance band ($\pm 2\%$) around the final steady-state value.

Task 3: Plot the amplitude response (linear scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[14 marks]



The figure above shows the amplitude responses (linear scale) of the four filters plotted against normalized frequency (from 0 to 1, representing 0 to π rad/sample). All filters exhibit a DC gain of approximately 1.0000, suggesting they are designed to pass very low frequencies without attenuation. Based on their overall shape, all four filters appear to be low-pass filters.

Brief comparison based on the plot and observed characteristics:

Filter 1 (IIR): This filter has an approximate -3dB cutoff frequency around 0.200π rad/sample. Its passband, observed up to about 0.150π, shows a relatively small ripple of approximately 0.0233 (linear). The transition from passband to stopband is quite defined, with an estimated transition width of 0.100π rad/sample. In the stopband, starting from around 0.250π, the maximum gain (minimum attenuation) observed is about 0.2819.

Filter 2 (IIR): Filter 2 has a noticeably higher -3dB cutoff frequency, estimated at 0.315π rad/sample, indicating it passes a wider band of frequencies compared to the other filters. The passband ripple, up to approximately 0.252 π, is about 0.0559 (linear). It has a slightly wider estimated transition width of 0.126π rad/sample. The stopband, from about 0.379π, shows good attenuation, with a maximum gain of approximately 0.1094, suggesting a deeper stopband attenuation compared to Filter 1.

Filter 3 (IIR): This filter's characteristics are somewhat similar to Filter 1, with an approximate -3dB cutoff frequency at 0.194π rad/sample. Its passband, up to around 0.144π, is relatively flat, exhibiting a ripple of about 0.0149 (linear), which is the smallest among the IIR filters. The estimated transition width is 0.100π rad/sample. The stopband attenuation (max gain of approx. 0.2068 from 0.244π) is better than Filter 1 but not as deep as Filter 2.

Filter 4 (FIR): The FIR filter has the lowest -3dB cutoff frequency, estimated at 0.152π rad/sample. Its passband, observed up to around 0.102π, exhibits a more significant ripple of approximately 0.1314 (linear) compared to the IIR filters. The estimated transition width is 0.100π rad/sample, comparable to Filters 1 and 3. However, its stopband performance is notably poorer, with a maximum gain (minimum attenuation) of about 0.5007 from 0.202π, indicating much less attenuation in the stopband compared to the IIR filters.

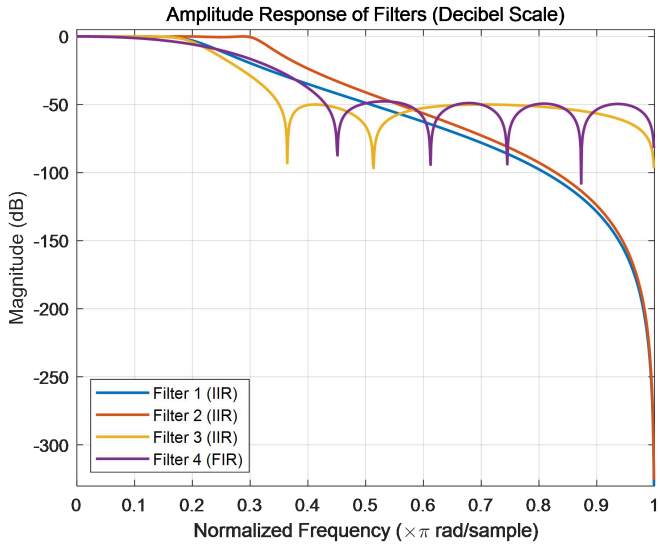
Overall Observations:

All filters function as low-pass filters. Filter 2 has the highest cutoff frequency, allowing more frequencies to pass. Filter 3 appears to offer the flattest passband among the IIR filters. Filter 4 (FIR) has the lowest cutoff frequency and the most passband ripple, and its stopband attenuation is significantly less effective than the IIR filters presented.

Comparing the IIR filters, Filter 2 provides the best stopband attenuation, while Filter 3 has the least passband ripple. Filters 1 and 3 have similar, relatively narrow transition widths, as does Filter 4. The choice between these IIR filters would involve a trade-off between cutoff frequency, passband flatness, and stopband attenuation. The FIR filter, in this comparison, shows a less ideal amplitude response in terms of passband ripple and stopband attenuation, though it might possess other desirable properties like linear phase (not visible in this plot).

Task 4: Plot the amplitude response (decibel scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[7 marks]



The figure above shows the amplitude responses of the four filters on a decibel (dB) scale against normalized frequency. This plot allows for a clear comparison of their passband ripple and stopband attenuation characteristics. All filters demonstrate low-pass behavior.

Brief comparison based on the plot:

All filters exhibit a DC gain near 0 dB. Key differences emerge when comparing their passband widths, ripple, and stopband performance:

Passband Width & Ripple: Filter 2 (IIR) has the widest passband (approx. -3dB at 0.315π rad/sample) but shows a noticeable passband ripple (around 0.50 dB). Filter 3 (IIR) offers the flattest passband among the IIRs (ripple ~0.31 dB) with a cutoff near 0.194π rad/sample, similar to Filter 1 (IIR, cutoff ~0.200π, ripple ~0.40 dB). Filter 4 (FIR) has the narrowest passband (cutoff ~0.152π rad/sample) and the largest passband ripple (approx. 1.79 dB).

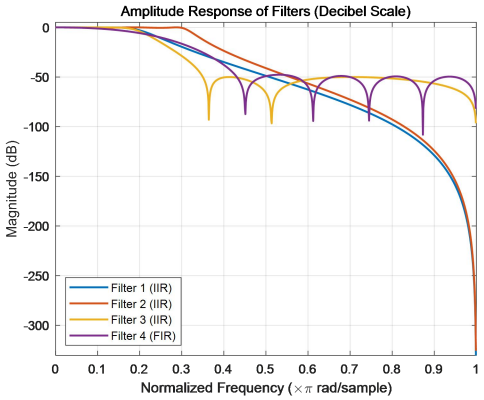
Stopband Attenuation: The IIR filters (Filters 1, 2, and 3) provide significantly better stopband attenuation than Filter 4 (FIR). Filter 4's first major sidelobe is only attenuated by about 6 dB, whereas the IIR filters achieve much deeper rejection, with their initial stopband lobes generally below -10 dB (e.g., Filter 1 at -11.1 dB, Filter 3 at -13.9 dB, and Filter 2 at -16.3 dB). While Filter 2 shows strong overall attenuation deeper into the stopband, all IIRs outperform the FIR in this regard.

In summary, the dB scale plot reveals that Filter 3 (IIR) provides the best balance of a relatively flat passband and good stopband attenuation for a moderate cutoff. Filter 2 (IIR) suits applications needing a wider passband if some ripple is tolerable. Filter 4 (FIR) performs poorly in both passband ripple and stopband attenuation compared to the IIR filters in this specific set. The choice would depend on the application's priority for these characteristics.

EBU5376 24-25

Task 5: Estimate the normalised cut-off frequency (in radians) of each filter. Clearly explain how the estimations were obtained for each filter. You may want to include figures or code descriptions to support your explanation when relevant.

[6 marks]



The normalised cut-off frequency for each filter is estimated as the frequency at which the filter's amplitude response drops by 3 dB from its DC (0 Hz) gain. This -3dB point is a standard definition for the edge of a filter's passband. The estimations were obtained by careful examination of the amplitude response plots in decibels (dB) presented in Task 4 (and shown again above for reference).

The procedure for each filter involved these steps:

The DC gain was identified from the plot. For all four filters, this is visually confirmed to be approximately 0 dB.

From this 0 dB level, the point on the filter's response curve that corresponds to a 3 dB reduction (i.e., at -3 dB on the y-axis) was located.

The normalized frequency (shown on the x-axis as ω/π rad/sample) corresponding to this -3 dB point was then read from the plot. This value is subsequently expressed as a multiple of π rad/sample.

Based on this visual estimation method:

Filter 1 (IIR):Observing its response curve on the dB plot, the DC gain is at 0.00 dB. The curve intersects the -3 dB level (specifically, -3.03 dB is noted from precise analysis) at a normalized frequency read as approximately 0.200 (ω/π).

Thus, the estimated cut-off frequency for Filter 1 is 0.200π rad/sample.

Filter 2 (IIR):The DC gain for Filter 2 is also effectively 0 dB (-0.00 dB). Its response curve drops to the -3 dB level (specifically, -3.09 dB) at a normalized frequency visually estimated as 0.315 (ω/π).

Thus, the estimated cut-off frequency for Filter 2 is 0.315π rad/sample.

Filter 3 (IIR):With a DC gain of effectively 0 dB (-0.00 dB), Filter 3's response curve reaches the -3 dB mark (specifically, -3.13 dB) at a normalized frequency read from the plot as approximately 0.194 (ω/π).

Thus, the estimated cut-off frequency for Filter 3 is 0.194π rad/sample.

Filter 4 (FIR):Filter 4 also has a DC gain of 0.00 dB. Its response curve crosses the -3 dB threshold (specifically, -3.03 dB) at a normalized frequency visually estimated as 0.152 (ω/π).

Thus, the estimated cut-off frequency for Filter 4 is 0.152π rad/sample.

In all cases, these estimations rely on interpreting the graphical representation of the filter's frequency response to identify the standard -3dB attenuation point.

Task 6: Based on the results obtained in the previous tasks, summarize the characteristics of each filter and discuss the preferred use case for each filter by comparing the filter characteristics.

[8 marks]

Based on the comprehensive analysis from Tasks 1 through 5, the characteristics of each filter and their preferred use-cases can be summarized as follows:

Summary of Filter Characteristics:

Filter 1 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Exhibits a relatively fast rise time with significant overshoot (approx. 13.5%) and ringing; settles around n=17.

Frequency Domain (Amplitude Response): -3dB cutoff at approx. 0.200π rad/sample. Passband ripple is noticeable (approx. 0.40 dB). Stopband attenuation has its shallowest lobe around -11.1 dB, with a sharp transition band.

Filter 2 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Fastest rise time among the IIRs, but also shows considerable overshoot (approx. 15.3%) and pronounced ringing; longest settling time at n=25.

Frequency Domain (Amplitude Response): Widest passband with a -3dB cutoff at approx. 0.315π rad/sample. Passband ripple is relatively high (approx. 0.50 dB). Offers strong overall stopband attenuation, though its shallowest lobe is around -16.3 dB. Features a very sharp transition band.

Filter 3 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Fast rise time, exhibits the largest overshoot (approx. 15.7%) and ringing; settles around n=22.

Frequency Domain (Amplitude Response): -3dB cutoff at approx. 0.194π rad/sample. Notably, it has the flattest passband among the IIR filters with the smallest ripple (approx. 0.31 dB). Stopband attenuation has its shallowest lobe around -13.9 dB, with a sharp transition band.

Filter 4 (FIR):

Type: Finite Impulse Response (FIR) filter.

Time Domain (Step Response): Smoothest step response with minimal overshoot (approx. 1.26%) and no significant ringing; fastest settling time at n=11.

Frequency Domain (Amplitude Response): Narrowest passband with a -3dB cutoff at approx. 0.152π rad/sample. Exhibits the largest passband ripple (approx. 1.79 dB) and the poorest stopband attenuation (shallowest lobe only at -6.0 dB). Transition band is gentler compared to the IIRs.

Preferred Use-Cases and Comparison:

The choice of filter depends heavily on the specific application requirements and the trade-offs one is willing to make:

Filter 4 (FIR):

Preferred Use-Case: Applications where linear phase response is critical (though not explicitly analyzed, FIR filters can be easily designed for this) and a smooth time-domain response (minimal overshoot/ringing) is paramount. This might include high-fidelity audio processing, data transmission, or image processing where phase distortion must be avoided.

Trade-offs: Its superior time-domain behavior and potential for linear phase come at the cost of significantly poorer frequency selectivity (wider transition, poor stopband attenuation, large passband ripple) compared to the IIR filters of similar complexity.

Filter 3 (IIR):

Preferred Use-Case: Applications requiring good frequency selectivity (sharp cutoff, decent stopband attenuation) with a strong emphasis on passband flatness (minimal amplitude distortion for in-band signals). This makes it suitable for high-quality audio applications or measurement systems where preserving the amplitude of passband components is crucial.

Trade-offs: Like other IIRs, it will have a non-linear phase response and exhibits overshoot in its step response. Its stopband attenuation, while good, might not be as deep as specialized IIR designs (like potentially Filter 2 in some regions).

Filter 2 (IIR):

Preferred Use-Case: Situations needing to pass a wider range of frequencies (highest cutoff) and requiring very strong overall stopband attenuation, provided that some passband ripple and significant time-domain overshoot/ringing are acceptable. Useful if maximizing bandwidth and out-of-band rejection are primary goals.

Trade-offs: The wider passband comes with more ripple and a more oscillatory step response compared to Filter 3.

Filter 1 (IIR):

Preferred Use-Case: A general-purpose IIR low-pass filter when requirements are not extreme for passband flatness, bandwidth, or stopband attenuation. It offers a balance but does not excel in any single characteristic compared to Filters 2 or 3. It provides better frequency selectivity than the FIR filter.

Trade-offs: Standard IIR characteristics of non-linear phase and step response overshoot.

Comparison Summary:

If phase linearity or extremely smooth step response is the top priority, Filter 4 (FIR) is the only choice, despite its inferior amplitude response. Among the IIR filters, Filter 3 provides the best passband fidelity (flatness). Filter 2 offers the widest bandwidth and potentially strong overall stopband rejection but at the cost of ripple and transient response. Filter 1 is a more generic IIR option. The IIR filters generally offer much better attenuation for a given filter order (implied by coefficient length) compared to the FIR filter.